

Sustainable Teaching Labs Stage 2

Prepared By

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TUTORIAL 5
Friday 9AM to 11AM

(Tanner & Kent, 2011)

Section 1. Problem Definition

In the past few years, there has been a noticeable increase in the number of engineering students at the University of Sydney. This presented the School of Aerospace, Mechanical & Mechatronics Engineering (AMME) with the challenge of finding enough space to accommodate for these students within teaching labs and workshops. Accessibility into these spaces has also been rather restricted towards students with different physical abilities and neurodiversity and this can be deemed detrimental to their academic progress. Additionally, the AMME labs and workshops have a relatively negative impact on the environment, as confirmed by academics, that “laboratories typically consume 4 to 5 times more energy than similarly sized commercial spaces” (Woolliams et al., 2005). This suggests that it is the faculty’s responsibility to ensure the sustainability of their teaching labs, as outlined by the University of Sydney’s Sustainability Strategy (USYD, 2020).

Section 2. Engineering Requirements

2.1 - All lab equipment must adhere to current 2024 safety protocols as outlined by the University of Sydney’s policies.

It is crucial to measure the quality of equipment based on its compliance with current safety standards. Ensuring that the equipment follows these protocols is not only legally required but also essential for the safety and well-being of everyone involved. Furthermore, the University's policy of Net Zero Initiative (USYD, 2023) drives this requirement.

2.2 - Reduce Scope 1 and 2 emissions and material wastage from the labs by 5 to 10% in the first year of implementing the new spaces/ideas.

It is required to significantly reduce both carbon emissions and waste by addressing the issues related to Arduino board wastage, tensile testing wastage, etc. This aligns with the shareholders’ (Industry, University and Government) goal of reaching the target of net-zero emissions through the deployment of affordable technologies and more (USYD, 2023).

2.3 - Budget allocated by the University must be followed as per the *Building Projects Approval and Management Policy 2014*.

Although no budget has yet been specified, when the project is approved, all work must strictly abide by the policies outlined in the *Building Project Approval and Management Policy 2014* and *Asset Financial Management Procedures 2019*, (Andrews, 2019). This requirement is directly taken from university policy and is therefore crucial to uphold.

2.4 - New spaces must be equipped with appropriate safety measures as per safety standards laid by the University of Sydney.

The teaching labs and workshops are to be fitted with a thermal fire detection and automatic alarm system and several break-glass points available in case of fire and non-fire emergencies (Hipsley, 2020). The building standards for learning spaces provided by USYD show that the upgraded labs will need to follow standards and policies to ensure the safety of all parties (Bolton et al., 2018).

2.5 - Special waste disposal unit must be installed to improve proper disposal of organic, hazardous and plastic waste in each teaching lab.

Development and implementation of a waste management solution to reduce, recycle and reuse waste is essential. This should align with the sustainability strategy of the University, to achieve zero waste to landfill by 2030, enable single use plastic free campuses, compost 80% of food waste, and achieve 60% recycling by 2025 (USYD, 2021).

2.6 - Workplace health and safety (WHS) policies and procedures must be implemented throughout the construction and operation of labs.

The requirement is derived from NSW law and the University WHS Policy 2016, which is to help prevent fatalities and serious injuries and enhance overall wellbeing of staff and students (Australian Government, 2020). Given the presence of potential hazardous material and equipment, prioritising WHS is crucial to mitigate risk and adherence to regulatory standards (USYD, 2016).

2.7 - Any renovations made must not compromise the heritage of the J07 building.

Under the university Heritage Asset Management Strategy, which aims to conserve any historical infrastructure present on campuses. This also aligns with the stakeholders' preferences which is to maintain the outside structural integrity of the building.

2.8 - All equipment and furniture must be accessible to students with diverse physical abilities and neurodiversity.

This may include the provision of ramps, lifts, and ADHD friendly features. This requirement is derived from the social contexts and client priorities concerning lab restrictions towards students with varying physical abilities and neurodiversity.

2.9 - High-quality demonstrators and tutors must be selected to teach in the session as per the standards set by the University of Sydney.

They should possess necessary teaching skills as well as adapt and learn new technologies quickly. The performance of the staff can be assessed through student feedback and surveys. This is derived from the University's Casual Employment Policy, Performance Planning & Development Policy 2012, Special Studies Program Policy 2015, and Staff & Affiliates Code of Conduct 2021.

2.10 - Increasing Functional Capacity of the current teaching labs and workspaces

This requirement is derived from the client priorities of accommodating to the increased student number. The social context of being fair to all students and providing them with sufficient space drives this need. According to academics, this can be fulfilled by implementing ideas like remote labs, collaborative work, flexible furniture and more (Van den Beemt et al., 2022).

Section 3. Idea Descriptions

3.1 - Idea 1: Expansion of Space

An expansion of lab space is proposed here, drawing inspiration from existing setups utilized by the Faculty of Agriculture and Environment and the Faculty of Health Sciences, which utilize demountable facilities located on Science Rd (Cass, 2023). To address the immediate need for additional lab capacity, temporary flat-pack laboratories can be deployed. These versatile structures can accommodate all labs and serve as a temporary solution while permanent spaces undergo renovation. The flexibility of these temporary labs enables easy relocation or repurposing once permanent upgrades are complete (MARC Tech, 2023). Moreover, using solar panels for energy and ensuring ramp/disabled access, fulfils the engineering requirements for sustainability and accessibility. This approach not only addresses the current capacity constraints but also provides the discussed long-term benefits.

3.2 - Idea 2: Virtual Simulations

Incorporating virtual labs or including simulations for certain activities will ensure a digital environment with reduction in waste production and physical space used for keeping equipment. It will also lay foundations for a more involved session and collaborative work while also providing

hands-on experiences. For example: Using Ansys software which is a Multiphysics engineering simulation software for product design (Ansys, 2024), testing and operation. Ansys' mechanical finite element analysis software will be especially useful for students in AMME2500 as it is used to simulate computer models of structures, electronics, or machine components. It can also be used to virtually test the models and each modelled component individually.

3.3 - Idea 3: Smart Laboratory Technologies

Since 2022, there have been constant improvements to the University campus technology to help reach its Net Zero goal such as HVAC systems, installation of motion sensor lighting, and Green Labs certification system, (TheGreenLab, n.d) (Australian Gov, 2023). Along with promoting sustainable practices among laboratory staff, students and researchers through different sustainability campaigns and policy. All these measures need to be improved and transferred over to the new improved lab along with additional biodigester and specialised waste bin or paper, organic waste and plastic (USYD, 2022). This aims to reduce waste to landfill and promote better reuse and recycling.

3.4 - Idea 4: Collaborative Work

To account for the increasing number of engineering students, it is proposed that more collaborative work takes place within the workshops and laboratories. This will result in less product/material wastage as each group would receive the same amount of product an individual student would usually receive. This aligns with USYD's sustainability strategy and goal to reduce overall waste, (USYD, 2020). There will also be movable furniture within the labs and workshop spaces to better accommodate for the students' collaboration and involvement. According to academics, it was found that individuals participating in teamwork are "80 percent more likely to report higher emotional well-being" (Middleton, 2023).

Section 4. Team Skills Summary

Samiksha Aveg

As a Software Engineering student with a keen interest in sustainable practices, Samiksha has volunteered for the 'Green Impact Auditor' program, which gave her extensive knowledge about the UN's initiatives for sustainability and USYD's Net Zero Initiative. She also refined her professional skills of verbal and written communication, analytical thinking, interpersonal skills, and project management through various group works.

Nicole Farah

During previous work experiences, design competitions and university projects, Nicole has developed the interpersonal and professional skills of communication, collaboration, time and project management and report writing. As a mechanical engineering student, she has participated in the first-year undergraduate workshops and AMME2500 laboratories and is hence well aware of the changes required to the spaces.

Minh Ho

Over the last 2 years at Uni as a biomedical engineering, Minh participated in different programs and events such as Humanitarian Hackathon, EWH Design competition in 2023. These are all client-based projects addressing real world problems. These helped him to improve his adaptability, communication, teamworking and project management skills. Moreover, his role as an ENGO mentor improved his networking abilities.

Jeremy Wells

Undertaking both a Civil Engineering and Environmental Science degree equips me with added experience regarding the Environmental sustainability aspect of the project. Civil engineers consider environmental factors when planning upgrades, such as energy efficiency, waste management, and sustainable construction practices, aligning to the University's net zero initiative.

Section 5. Decision Matrix

The following decision matrix was produced to determine and rank the feasibility and success of each proposed idea. The weightings and scores selected for each criterion and idea were a result of thorough research and group collaboration. It was determined that the requirements that were driven by legal and environmental context weighed the most as the ideas should be legally compliant and environmentally friendly. The client priorities were weighed with respect to the research conducted in stage 1 of this project, giving their priority to increase lab capacity the highest weighting. It was also decided that the research and communication skills would be the most relevant for all the ideas as this project will require a large amount of groupwork for success.

Furthermore, the criteria for each idea were scored by the group as a whole, using a large amount of collaboration and discussion. The *Smart Laboratory Technologies* idea was deemed the most successful, as its scores were high and consistent for most of the criteria making it rank the overall highest. The ideas of *Virtual Simulations* and *Collaborative Work* scored high in some requirements but lower in other potentially more important criteria. This inconsistency meant a lowered overall score; however, they could still be essential to the final solution of the project. Ultimately, after thorough decision making, the *Expansion of Space* idea scored the lowest and this suggested that it may not be as feasible as the rest. Construction to existing spaces may disrupt the education progression of students and may also lead to excessive noise and dust pollution in the area. Hence, this idea scored low especially in the environmental context. Overall, the decisions made throughout the matrix allowed the group to identify the practicality of each proposed idea.

Engineering Requirements	Criteria	Weighting	Expansion of Space Score	Weighted score	Virtual Simulation Score	Weighted score	Smart Laboratory Technologies Score	Weighted score	Collaborative Work Score	Weighted score
Client priority	Sustainable labs	6	2	12	4	24	7	42	4	24
	Increased capacity of the teaching labs and workspaces	7	7	49	4	28	7	49	7	49
	Accessibility for different physical or neurodiverse students	5	2	10	5	25	7	35	5	25
Problem definitions	Lack of accessibility for students with diverse physical abilities and neurodiversity	5	3	15	6	30	8	40	5	25
	Lack of Sustainability of the teaching labs & workspaces	6	5	30	8	48	8	48	5	30
	Increase in enrolled students	8	5	40	4	32	5	40	8	64
	Lack of functional capacity in workshops and labs	8	5	40	4	32	9	72	5	40
Social context	Enhance accessibility for students with diverse physical abilities and neurodiversity	6	4	24	6	36	8	48	5	30
	Maintain wellbeing of students and teachers in class	9	1	9	5	45	8	72	6	54
Legal Context	According to WHS Government Regulations, maintain wellbeing of staff & students	9	2	18	5	45	8	72	8	72
	Maintaining heritage of the building	7	4	28	10	70	7	49	10	70
	Faculty to strictly follow the budget allocated by the University	8	2	16	8	64	7	56	9	72
Policy context	Accessibility for students with diverse physical abilities and neurodiversity	6	4	24	6	36	8	48	7	42
	Maintain WHS policies of individuals during construction and teaching	9	6	54	5	45	3	27	7	63
	Maintaining heritage of the building as per Uni policy	8	5	40	10	80	8	64	10	80
	New spaces equipped with safety measures as per standards laid by USYD	8	5	40	7	56	8	64	3	24
	Finding high-quality demonstrators as per UniSyd standards	9	1	9	6	54	5	45	6	54
Environmental context	Special waste disposal units should be used	7	2	14	7	49	5	35	3	21
	Reduction of emissions & waste	8	1	8	8	64	8	64	7	56
	Equipments in line with safety & sustainability protocols	6	2	12	7	42	8	48	4	24
	Follow Net Zero Initiative	7	2	14	9	63	8	56	6	42
Team Skills Base	Extensive researching skills	8	6	48	7	56	7	56	7	56
	Teamwork/Collaboration	8	6	48	8	64	9	72	10	80
	Communication	8	6	48	7	56	9	72	9	72
	Construction skills	6	8	48	6	36	6	36	5	30
	Professional Skills	6	5	30	5	30	7	42	7	42
	Adaptability	6	6	36	8	48	7	42	6	36
SCORE				764		1258		1394		1277

Table 1. Decision Matrix for Proposed Ideas

Section 6. References

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